

Sound Pressure Level [dB(A)] S4000 Diesel Gensets

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Genset Surface Noise [dB(A)] without radiator

Type		Engine	Gross Power [KWe]	L _{WA} 1m [dB(A)]	L _{pA} 1m [dB(A)]
MTU 12V4000 DS2250	3E	12V4000G34F		-	-
MTU 12V4000 DS2250	3D	12V4000G94F		-	-
MTU 20V4000 DS3600	3B	20V4000G44F	*	127	107,2
MTU 20V4000 DS3600	3D	20V4000G94F	2870	127	107,2
MTU 20V4000 DS4000	3E	20V4000G44LF	*	127	107,4
MTU 20V4000 DS4000	3D	20V4000G94LF	3090	127	107,4

Tolerance: + 2dB(A) for A-rated sound level

*As reference, measurement was done for Standby Power

Genset Surface Noise [dB(A)] with radiator

Type		Engine	Net Power [KWe]	L _{WA} 1m [dB(A)]	L _{pA} 1m [dB(A)]
MTU 12V4000 DS2250	3E	12V4000G34F	1620	124	103
MTU 12V4000 DS2250	3D	12V4000G94F	1785	124	103
MTU 16V4000 DS2750	3D	16V4000G94F	2387	125	103
MTU 16V4000 DS2750	3E	16V4000G34F	2170	125	103
MTU 20V4000 DS3600	3D	20V4000G94F	3088	133	110

Tolerance: + 2dB(A) for A-rated sound level



12V 4000 G94F Genset

AIRBORNE NOISE ANALYSIS
1785 kWel / 1500 rpm

TCE Golbach

Drawing No.: 737 912e
Date: 15.10.2018

Genset Surface Noise Analysis - 1/3-Octave

ENGINE TYPE:	12V 4000 G94F	ENGINE NO.:	547100484
GENERATOR:	Leroy Somer	TYPE:	LSA 52.3 S7/4p
POWER / SPEED:	1785 kWel / 1500 rpm	TEST CELL:	Ruhstorf PS A
FAN COOLER:	Bearward - 12 blade	DATE MEASURED:	19.09.18
TURBOCHARGER:	4 x GT 5008		
INTAKE AIR OPENING:	Paper filter without housing		
MEASURING DISTANCE:	1 m		
MEASURING SURFACE DIMENSION:	21.2 dB		
NO. OF MEASURING POINTS:	11		
SOUND PROPAGATION:	Free-field		
MEASUREMENT STANDARD:	similar to ISO 8528-10		
TOLERANCE:	+5 dB for single 1/3-octave band, +2 dB(A) for total A-weighted level.		

Energy mean sound pressure levels of the airborne noise that is emitted by the genset surface.
For project purposes only. Incl. noise of radiator with fan ventilating in the test cell.

Energy mean free-field levels Total: L = 106.4 dB LA = 103.2 dB(A)

f [Hz]	12.5	16	20	25	31.5	40	50	63	80	100	125	160	200	250	315	400	500
LpF [dB]				74.8	72.2	81.4	77.5	83.4	93.1	93.8	94.6	89.7	96.9	93.7	95.2	93.8	94.3

f [Hz]	630	800	1k	1.25k	1.6k	2k	2.5k	3.15k	4k	5k	6.3k	8k	10k	12.5k	16k	20k
LpF [dB]	93.3	93.5	93.1	91.2	94.7	90.9	89.8	88.6	87.9	87.2	84.9	83.8	83.6	82.4	95.5	82.4

